SUPPLEMENTARY MATERIAL FOR "WHERE VIOLENCE PERSISTS: LATENT SPATIAL FIELDS FOR THE RESIDUAL RISK OF ARMED CONFLICT AND SOCIAL UNREST"

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This supplement contains the detailed tables and figures referred to in the main paper. Equation, section and table numbers without the prefix S refer to the main paper.

S1. Methods.

S2. Simulation studies.

S3. Application.

S4. Additional results.

TABLE S1

Latent blocks of the latent Gaussian model of Section 3.1 of the main paper, their precision matrices and hyperparameters.

Block	Dimension	Precision	Hyperparameters
fixed effects $\boldsymbol{\beta}, \delta, \boldsymbol{\gamma}, \boldsymbol{\alpha}$	p	$10^{-4}\boldsymbol{I}$	–
country effects $\boldsymbol{u}$	n_c	$\sigma_u^{-2}\boldsymbol{I}$	σ_u
spatial field $\boldsymbol{w}$	n	$\boldsymbol{Q}_s(\rho_w, \sigma_w)$, SPDE precision	ρ_w, σ_w
cell effects $\boldsymbol{e}$	n	$\sigma_e^{-2}\boldsymbol{I}$	σ_e
SVC fields $\boldsymbol{\eta}_k$	n each	$\boldsymbol{Q}_s(\rho_k, \sigma_k)$	ρ_k, σ_k
space-time field $\boldsymbol{v}$	nT_v	$\boldsymbol{Q}_{\mathrm{AR}}(\phi) \otimes \boldsymbol{Q}_s(\rho_v, \sigma_v)$	ρ_v, σ_v, ϕ

TABLE S2

Boundary treatments on a 60×30 *lattice of* 0.5° *cells with a sea strait of 2, 4 or 6 cells (about 110, 220, 330 km): marginal variance of a coastal and of an inland land cell (nominal 1) and correlation between the two coastal cells facing each other across the strait, from 4,000 exact draws with* $\rho = 400$ *km. "Land only" removes the strait from the lattice; "sea = land" keeps it as ordinary cells without data.*

Strait		Land only	Sea = land	$f = 0.1$	$f = 0.2$	$f = 0.5$
2 cells	var. coast	1.98	1.07	1.80	1.67	1.29
	var. inland	1.10	1.06	1.09	1.11	1.09
	corr. across	-0.01	0.49	-0.03	0.01	0.16
4 cells	var. coast	2.00	1.08	1.77	1.69	1.29
	var. inland	1.06	1.05	1.09	1.09	1.09
	corr. across	0.00	0.28	0.01	0.01	0.04
6 cells	var. coast	2.02	1.04	1.76	1.66	1.26
	var. inland	1.06	1.05	1.08	1.07	1.07
	corr. across	-0.02	0.14	0.01	0.01	0.01

TABLE S3

S1: residual diagnostics and recovery of the field (mean and SD over 20 replicates).

Quantity	Mean	SD
Moran's I of cell residuals, XGBoost as in the original	0.360	0.050
Moran's I of cell residuals, cross-fitted XGBoost	0.637	0.067
$\mathrm{cor}(\hat{w}, w)$, hybrid with $u + w$	0.849	0.044
$\hat{\rho}_w$ (km) [truth 400]	409	50
$\hat{\sigma}_w$ [truth 1]	0.829	0.069

TABLE S4

S2: estimation of the effect α of a national indicator (30 replicates per scenario). Coverage of nominal 95% intervals, and the proportion of replicates in which the interval excludes zero. †Least-squares slope of SHAP values on Z_0, with the 95% block-bootstrap percentile interval and interquartile range.

Estimator	Mean	Emp. SD	Mean SE	Coverage	Reject $\alpha = 0$
No confounding, $\alpha = 0$					
Logistic regression	-0.007	0.107	0.016	0.23	0.77
+ country effects u	0.016	0.109	0.123	0.97	0.03
+ spatial field w	0.002	0.065	0.052	0.93	0.07
+ u + w	0.000	0.063	0.081	1.00	0.00
spatial+ with u + w	0.002	0.072	0.082	0.97	0.03
XGBoost SHAP slope†	-0.006	0.063	0.044	0.80	0.20
Confounding, $\alpha = 0$					
Logistic regression	0.465	0.115	0.018	0.00	1.00
+ country effects u	0.493	0.098	0.078	0.00	1.00
+ spatial field w	0.200	0.098	0.056	0.30	0.70
+ u + w	0.200	0.095	0.077	0.33	0.67
spatial+ with u + w	0.153	0.092	0.079	0.60	0.40
XGBoost SHAP slope†	0.332	0.132	0.056	0.00	1.00
Confounding, $\alpha = -0.3$					
Logistic regression	0.219	0.114	0.017	0.00	1.00
+ country effects u	0.235	0.104	0.079	0.00	0.80
+ spatial field w	-0.090	0.100	0.056	0.20	0.40
+ u + w	-0.091	0.098	0.077	0.37	0.33
spatial+ with u + w	-0.130	0.095	0.078	0.50	0.40
XGBoost SHAP slope†	0.138	0.091	0.049	0.00	0.87

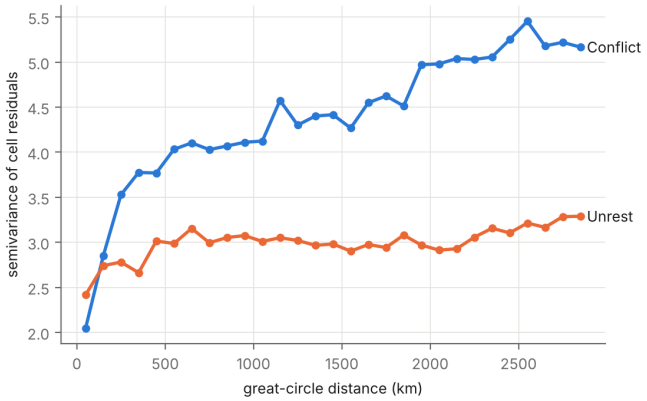


FIG S1. *Empirical semivariograms of the standardised test-period cell residuals, against great-circle distance.*

TABLE S5

S4: estimation of biome-specific temperature effects (30 replicates). "Detects sign": proportion of replicates in which the interval excludes zero on the correct side. ‡Median SHAP value of raw temperature within the biome, with the interquartile range of the bootstrap replicates, as in the original analysis; it estimates neither γ_b nor $\bar{\beta}_b$.

Estimator / biome	Mean est.	Cov. γ_b	Cov. $\bar{\beta}_b$	Detects sign
Biome slopes, no latent effects				
biome 1 ($\gamma = -0.3$)	-0.273	0.57	0.77	1.00
biome 2 ($\gamma = -0.1$)	-0.083	0.57	0.83	0.60
biome 3 ($\gamma = +0.1$)	0.108	0.53	0.70	0.73
biome 4 ($\gamma = +0.3$)	0.307	0.50	0.53	1.00
Biome slopes + u + w				
biome 1 ($\gamma = -0.3$)	-0.286	0.53	0.87	1.00
biome 2 ($\gamma = -0.1$)	-0.082	0.57	0.90	0.67
biome 3 ($\gamma = +0.1$)	0.119	0.60	0.73	0.73
biome 4 ($\gamma = +0.3$)	0.319	0.53	0.87	1.00
SVC: biome means + η + u + w				
biome 1 ($\gamma = -0.3$)	-0.277	0.97	1.00	0.93
biome 2 ($\gamma = -0.1$)	-0.077	0.93	1.00	0.23
biome 3 ($\gamma = +0.1$)	0.111	0.90	1.00	0.33
biome 4 ($\gamma = +0.3$)	0.309	0.83	1.00	0.97
XGBoost SHAP slope (targets $\gamma_b/2$)				
biome 1 ($\gamma = -0.3$)	-0.092	0.03	0.03	0.60
biome 2 ($\gamma = -0.1$)	-0.026	0.47	0.53	0.23
biome 3 ($\gamma = +0.1$)	0.048	0.47	0.40	0.37
biome 4 ($\gamma = +0.3$)	0.091	0.03	0.00	0.60
XGBoost SHAP slope, interactions credited				
biome 1 ($\gamma = -0.3$)	-0.191	0.27	0.40	0.83
biome 2 ($\gamma = -0.1$)	-0.056	0.63	0.63	0.30
biome 3 ($\gamma = +0.1$)	0.089	0.57	0.73	0.53
biome 4 ($\gamma = +0.3$)	0.176	0.40	0.30	0.80
XGBoost median SHAP, raw temperature‡				
biome 1 ($\gamma = -0.3$)	-0.052	0.03	0.07	0.57
biome 2 ($\gamma = -0.1$)	-0.151	0.13	0.13	0.97
biome 3 ($\gamma = +0.1$)	-0.150	0.00	0.00	0.00
biome 4 ($\gamma = +0.3$)	-0.043	0.00	0.00	0.20

TABLE S6

S4: recovery of the spatially varying coefficient $\beta(s)$ (mean and SD over 30 replicates).

Quantity	Mean	SD
$\mathrm{cor}\{\hat{\beta}(s), \beta(s)\}$, SVC	0.918	0.036
RMSE of $\hat{\beta}(s)$, SVC	0.106	0.010
RMSE of $\hat{\beta}(s)$, biome slopes only	0.147	0.020
$\hat{\sigma}_\eta$ [truth 0.15]	0.133	0.025
$\hat{\rho}_\eta$ (km) [truth 600]	564	261

TABLE S7

Test-period (2023–2024) residual diagnostics, Africa and the Middle East, for every model on identical test rows: Moran's I of the standardised cell residuals (permutation p-value from 999 permutations, bounded below by 0.001) and the residual correlation between neighbouring cells in the same country and across a border.

Outcome	Model	Moran's I	p	Neighbour residual corr.	
				within country	across border
Conflict	XGBoost, corrected	0.424	≤0.001	0.449	0.214
Conflict	XGBoost + cell event rate	0.295	≤0.001	0.301	0.229
Conflict	XGBoost, cross-fitted by country	0.539	≤0.001	0.585	0.285
Conflict	Cross-fitted XGBoost + u + iid cell effects	0.450	≤0.001	0.476	0.115
Conflict	Cross-fitted XGBoost + $u + w$	0.449	≤0.001	0.470	0.136
Conflict	xgb_dedup	0.354	≤0.001	0.372	0.226
Unrest	XGBoost, corrected	0.148	≤0.001	0.161	0.081
Unrest	XGBoost + cell event rate	0.253	≤0.001	0.277	0.106
Unrest	XGBoost, cross-fitted by country	0.258	≤0.001	0.287	0.097
Unrest	Cross-fitted XGBoost + u + iid cell effects	0.292	≤0.001	0.333	-0.013
Unrest	Cross-fitted XGBoost + $u + w$	0.287	≤0.001	0.327	-0.016
Unrest	xgb_dedup	0.156	≤0.001	0.166	0.117

TABLE S8

Fitted hyperparameters of the hybrid models, Africa and the Middle East: range ρ_w and standard deviation σ_w of the field, standard deviation of the independent cell effects, and standard deviation σ_u of the country effects.

Outcome	Latent terms	$\hat{\rho}_w$ (km)	$\hat{\sigma}_w$	$\hat{\sigma}_{\text{cell}}$	$\hat{\sigma}_u$
Conflict	$u + w$	263	1.48	–	0.76
Conflict	u + iid cell effects	–	–	1.30	1.37
Unrest	$u + w$	65	1.25	–	1.01
Unrest	u + iid cell effects	–	–	1.15	1.06

TABLE S9

Posterior summaries of the field w of the hybrid model averaged over regions: mean, standard deviation and posterior probability that the regional average is positive. The regions were chosen before the fit, as known conflict zones and regions expected to be quiet. The last two rows give the share of all cells whose posterior probability of $w(s) > 0$ is above 0.95 or below 0.05.

Region	Longitude	Latitude	Cells	Conflict		Unrest	
				mean (SD)	$P(>0)$	mean (SD)	$P(>0)$
Eastern DRC	27 to 30.5	-5 to 2	94	1.49 (0.18)	1.00	0.52 (0.11)	1.00
Northern Mozambique	38 to 41	-14 to -10	39	1.44 (0.29)	1.00	0.14 (0.20)	0.76
Syria and Iraq	36 to 46	32 to 38	240	2.41 (0.21)	1.00	-0.03 (0.10)	0.36
Yemen	43 to 49	13 to 17	84	1.95 (0.37)	1.00	0.04 (0.16)	0.61
Lake Chad	12 to 16	10 to 14	64	1.66 (0.19)	1.00	-0.19 (0.13)	0.07
Central Sahel	-5 to 5	11 to 17	240	1.56 (0.18)	1.00	-0.03 (0.09)	0.39
Horn of Africa	38 to 48	3 to 10	277	0.41 (0.21)	0.97	-0.04 (0.09)	0.32
Sudan	23 to 35	10 to 16	288	-0.06 (0.23)	0.40	0.09 (0.09)	0.83
Arabian interior	42 to 55	18 to 27	423	-0.61 (0.30)	0.02	-0.03 (0.09)	0.38
Western Congo basin	14 to 24	-5 to 3	320	-1.10 (0.22)	0.00	-0.13 (0.09)	0.07
Southern Africa	18 to 32	-34 to -22	619	-0.45 (0.26)	0.04	0.00 (0.08)	0.53
Share of cells with $P\{w(s) > 0\} > 0.95$			13719	0.189		0.116	
Share of cells with $P\{w(s) > 0\} < 0.05$			13719	0.030		0.026	

TABLE S10

Dynamic experiments for the hybrid models, Africa and the Middle East. In all of them the offset is the country cross-fitted XGBoost prediction. (a) Oracle: the latent terms are fitted to the 2023–2024 test weeks themselves, with the offset fixed, to show how much residual structure the test period contains and whether it is smooth in space; the correlation compares this field with the field fitted on 2018–2022, and Moran's I is computed on the in-sample cell residuals. (b) Scores on the 2024 test weeks only (713,388 rows per outcome), for models fitted on 2018–2022 and for models whose offset and latent terms are retrained on 2018–2023 (training years in parentheses); differences in log score are relative to the corrected XGBoost model, with SEs clustered by country. (c, d) Hyperparameters of the space-time hybrids, in which an AR(1) $\otimes$ Matérn field v on 1° cells replaces the static field w; ϕ is the AR(1) coefficient per time step, and the number of steps in parentheses is the number that falls in the training period. (e) Quarterly update: the latent terms are re-estimated at the start of each quarter of the test period with all data observed so far, and the hyperparameters are kept at their 2018–2022 values. Out-of-sample scores of the models in (c), (d) and (e) are in the forecasting comparison of Section 5 of the main paper.

Quantity	Conflict	Unrest
(a) Oracle: latent terms fitted to the 2023–2024 test weeks, offset fixed		
log marginal likelihood, $u + w$ minus u + iid cell effects	1045.4	69.0
$\hat{\rho}_w$ (km)	374	66
$\hat{\sigma}_w$	1.49	1.08
cor(oracle $\hat{w}$, $\hat{w}$ fitted on 2018–2022)	0.79	0.69
Moran's I of cell residuals, offset alone	0.539	0.258
Moran's I of cell residuals, u + iid cell effects	0.346	0.154
Moran's I of cell residuals, $u + w$	-0.119	0.003
(b) 2024 test weeks only: AUC-PR; Δ log score $\times 10^3$ (SE)		
XGBoost, corrected (2018–2022)	0.522; reference	0.553; reference
XGBoost + cell event rate (2018–2022)	0.533; -2.78 (1.42)	0.562; -2.34 (0.96)
Offset + $u + w$ (2018–2022)	0.520; -0.67 (1.22)	0.557; -1.73 (0.67)
Cross-fitted XGBoost (2018–2023)	0.487; 6.27 (1.46)	0.526; 3.59 (1.11)
Offset + u + iid cell effects (2018–2023)	0.535; -1.85 (0.76)	0.572; -3.24 (0.79)
Offset + $u + w$ (2018–2023)	0.535; -2.17 (0.78)	0.572; -3.27 (0.79)
$\hat{\rho}_w$ (km), $u + w$ (2018–2023)	248	66
Test rows (2024)	713,388	713,388
(c) Space-time hybrid, offset + u + cell effects + v, 13-week steps on 1° cells		
$\hat{\rho}_v$ (km)	779	996
$\hat{\sigma}_v$	1.39	0.62
$\hat{\phi}$ (per 13 weeks)	0.986	0.949
$\hat{\sigma}_e$	0.81	1.07
$\hat{\sigma}_u$	0.79	0.96
Time steps (training)	29 (21)	29 (21)
Fit time (h)	2.2	2.3
(d) Space-time hybrid, offset + u + cell effects + v, 26-week steps on 1° cells		
$\hat{\rho}_v$ (km)	731	891
$\hat{\sigma}_v$	1.33	0.63
$\hat{\phi}$ (per 26 weeks)	0.974	0.943
$\hat{\sigma}_e$	0.80	1.06
$\hat{\sigma}_u$	0.79	0.94
Time steps (training)	15 (11)	15 (11)
Fit time (h)	0.4	0.3
(e) Quarterly update: hyperparameters fixed at their 2018–2022 values		
$\hat{\rho}_w$ (km)	263	65
$\hat{\sigma}_w$	1.48	1.25
Number of updates	8	8

TABLE S11

S5: full results for the escalation effect κ of recent unrest on conflict, by scenario and estimator, over 20 replicates per scenario. Mean $\hat{\kappa}$, its empirical SD across replicates and the mean of the estimated SEs; coverage of the nominal 95% intervals for κ with its Monte Carlo SE; proportion of replicates whose interval excludes zero; mean estimate of the own-lag coefficient d_C (truth 0.3); mean and SD of the estimated loading λ of the shared space-time field (truth 0.8, joint model only); and mean log score, AUC-PR and onset AUC-PR of the conflict forecasts over the 26 test weeks. Onset rows are those of cells without a conflict event in the previous 52 weeks.

Estimator	Mean $\hat{\kappa}$	Emp. SD	Mean SE	Coverage (MC SE)	Reject $\kappa = 0$	Mean $\hat{d}_c$	Mean $\hat{\lambda}$ (SD)	Log score	AUC-PR	Onset AUC-PR
No escalation, $\kappa = 0$										
Logistic regression	0.695	0.193	0.129	0.05 (0.05)	0.95	1.54	–	0.1372	0.063	0.022
Static field w_c	0.426	0.161	0.135	0.15 (0.08)	0.85	0.82	–	0.1335	0.090	0.037
Static field w_c + space-time field	0.269	0.158	0.140	0.45 (0.11)	0.55	0.38	–	0.1341	0.090	0.034
Joint model, shared space-time field	0.042	0.166	0.138	0.90 (0.07)	0.10	0.38	0.83 (0.14)	0.1343	0.088	0.033
Escalation, $\kappa = 0.4$										
Logistic regression	1.026	0.163	0.116	0.00 (0.00)	1.00	1.52	–	0.1430	0.077	0.025
Static field w_c	0.794	0.128	0.122	0.10 (0.07)	1.00	0.80	–	0.1392	0.105	0.044
Static field w_c + space-time field	0.659	0.108	0.127	0.50 (0.11)	1.00	0.37	–	0.1399	0.105	0.045
Joint model, shared space-time field	0.434	0.111	0.126	0.95 (0.05)	0.95	0.36	0.83 (0.15)	0.1401	0.103	0.042

TABLE S12

Joint model of conflict and unrest, by region (Section 5 of the main paper): all hyperparameters and lag coefficients (posterior mean, with SE in parentheses) of the two univariate conflict models and of the joint model, fitted on 2018–2022. Subscripts C and U denote conflict and unrest. In the univariate model with $u + e + w + v$, the space-time field v belongs to conflict alone; in the joint model it is shared, with loading λ for unrest. Histories are $\log(1 +$ events in the previous four weeks$)$; neighbour histories are means over the rook neighbours, standardised on the training rows. The log marginal likelihood of the joint model includes the unrest data and is not comparable with those of the univariate models. Test scores are for 2023–2024 in the same region.

Quantity	Conflict only $u + e + w$	Conflict only $u + e + w + v$	Joint shared v
West Africa and Sahel			
2,505 cells, 22 countries, 906,810 rows per outcome			
v on 641 1° cells, 28 steps of 13 weeks			
Hyperparameters			
$\sigma_{u,\mathrm{C}}$	0.69	0.67	0.69
$\sigma_{e,\mathrm{C}}$	0.65	0.64	0.66
$\rho_{w,\mathrm{C}}$ (km)	514	508	453
$\sigma_{w,\mathrm{C}}$	1.70	1.74	1.39
$\sigma_{u,\mathrm{U}}$	–	–	0.53
$\sigma_{e,\mathrm{U}}$	–	–	1.34
$\rho_{w,\mathrm{U}}$ (km)	–	–	369
$\sigma_{w,\mathrm{U}}$	–	–	0.21
ρ_v (km)	–	423	499
σ_v	–	0.69	1.01
ϕ (per step)	–	0.905	0.952
λ	–	–	0.868
Conflict equation			
δ_{C}: own history	0.575 (0.016)	0.444 (0.017)	0.459 (0.017)
κ: unrest history	0.138 (0.031)	0.131 (0.032)	0.044 (0.032)
$\delta_{\mathrm{C}}^{\mathrm{nb}}$: neighbours' conflict	0.129 (0.007)	0.051 (0.008)	0.061 (0.008)
κ^{nb}: neighbours' unrest	0.005 (0.008)	0.003 (0.008)	−0.020 (0.008)
Unrest equation			
δ_{U}: own history	–	–	0.351 (0.029)
κ_{U}: conflict history	–	–	0.066 (0.031)
$\delta_{\mathrm{U}}^{\mathrm{nb}}$: neighbours' unrest	–	–	0.017 (0.009)
$\kappa_{\mathrm{U}}^{\mathrm{nb}}$: neighbours' conflict	–	–	−0.042 (0.013)
Fit			
Log marginal likelihood	−49530.1	−49080.6	−76591.8
Fit time (h)	0.1	0.7	4.5
Test AUC-PR, conflict	0.470	0.471	0.466
Test log score, conflict	0.1020	0.1026	0.1031
Test AUC-PR, unrest	–	–	0.289
Test log score, unrest	–	–	0.0490
Africa and Middle East: [TODO: running]			